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Laundry starch/sizing/fabric finish products: HMTc Standard

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CORRESPONDENCE

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for laundry starch/sizing/fabric finish products in Laundry and Fabric-Contact Home Products, evaluated on total heavy-metal content of the finished product as placed on the market (as-sold consumer-chemical product-content basis, mg/kg of product). 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; no sovereign sets a contaminant maximum for a laundry or fabric-care product, so each toxic-panel cell is a disclosed read-across from the nearest applicable sovereign consumer-chemical maximum level in the product's native as-sold content basis, and each remaining unregulated cell is a control with a published method and escalation rule; never occurrence-derived)

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	2,000	human-grade read-across (DE)
Cadmium (Cd)	Tier 1	100	human-grade read-across (DE)

Total arsenic (tAs)	Tier 2	500	human-grade read-across (DE)
Inorganic arsenic (iAs)	Tier 1	n/m	not material in this matrix
Mercury, total (tHg)	Tier 1	100	human-grade read-across (DE)
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	10,000	human-grade read-across (South Korea)
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	2,000	DE — human-food analog
Cadmium (Cd)	100	DE — human-food analog
Total arsenic (tAs)	500	DE — human-food analog

Inorganic arsenic (iAs)	n/m	Inorganic arsenic is the toxic fraction of total arsenic and a species of it (iAs <= tAs). No jurisdiction sets an inorganic-arsenic limit for a laundry product, a fabric-care product, a detergent or a cosmetic: every consumer-chemical arsenic instrument in the register regulates TOTAL arsenic (China NMPA 2015 Table 2, China GB 14930.1-2022, Korea MFDS Notice 2025-63 and 2024-9, Germany BVL 2017, Health Canada, the ASEAN Annex, and EU REACH Annex XVII entry 72 Appendix 12), and no sovereign speciates arsenic for any product in this category. The inorganic fraction is therefore governed by, and bounded by, the total-arsenic ceiling published on the tAs cell of this row. The determination is a governance statement about which cell carries the ceiling, not an assertion that inorganic arsenic is absent, and it is rebuttable: the only routed inorganic-arsenic measurement anywhere in this category's evidence base is zhang2026-cotton-arsenic--amf-calcium, which speciates inorganic arsenic in COTTON FIBRE as a soil-uptake pathway in the textile substrate, not as a contaminant of a product applied to the textile, so it does not rebut this determination for the product itself. The starch base is an agricultural feedstock, so crop-uptake metals travel into the product with the raw material. Rice starch in particular is drawn from a recognised arsenic accumulator, which is why total arsenic and the reflex inorganic-arsenic speciation carry additional weight on this row and why the declared starch botanical origin is recorded with every arsenic result.
Mercury, total (tHg)	100	DE — human-food analog

Methylmercury (MeHg)	n/m	Methylmercury is a species of total mercury (MeHg <= tHg) formed by microbial methylation in aquatic food webs and concentrated by dietary bioaccumulation in fish muscle. A surfactant, quaternary--ammonium, silicone, starch or mineral laundry formulation has no methylation pathway, no aquatic food-web input and no dietary mercury source, and no routed source in this category's evidence base measures or speciates methylmercury in any product form — the sector standards and the product surveys measure TOTAL mercury. Any mercury present is governed by, and bounded by, the total-mercury ceiling published on the tHg cell of this row. No jurisdiction sets a methylmercury limit for a laundry, fabric-care, detergent or cosmetic product; sovereign methylmercury maximum levels are scoped to fish and seafood.
Nickel (Ni)	10,000	South Korea — human-food analog
Tin (Sn)	control	No sovereign sets a total- or inorganic-tin contaminant maximum for a laundry or fabric-care product, and no near-neighbour read-across is supportable. Instruments checked: China GB 14930.1-2022 (the only sovereign detergent contaminant standard in the register; scope is Type A detergents used directly for food cleaning and Type B detergents for tableware and food-contact tools, equipment, packaging and containers, and it sets a total-arsenic ceiling only); China NMPA 2015 Safety and Technical Standards for Cosmetics Table 2 (lead, arsenic, mercury, cadmium); Korea MFDS Notice 2025-63 and its predecessor Notice 2024-9 (lead, cadmium, nickel, antimony, arsenic, mercury); Germany BVL 2017 technically-avoidable heavy metals in cosmetics (lead, cadmium, mercury, arsenic, antimony); Health Canada cosmetic heavy-metal impurity limits; the ASEAN Cosmetic Directive Annex (2019) and the Malaysia NPRA, Singapore HSA, Indonesia BPOM, Taiwan MoHW and New Zealand cosmetic group-standard equivalents; EAEU TR CU 009/2011; EU REACH Annex XVII entry 20 (organotin), entry 23 (cadmium in listed polymers and paints), entry 47 as amended by Regulation (EU) 301/2014 (Cr(VI) in skin-contact leather), entry 72 with Appendix 12 as amended by Regulation (EU) 2018/1513 (lead, cadmium, arsenic and Cr(VI) in consumer clothing and skin-contact textiles) and entry 75 under Regulation (EU) 2020/2081 (tattoo and permanent-make-up mixtures); EU Directive 2009/48/EC toy-safety migration limits; EU RoHS 2011/65/EU; and the dietary contaminant regimes (EU

Aluminium (Al)	control	No jurisdiction sets an aluminium maximum for any consumer-chemical product, and no near-neighbour read-across is supportable. Instruments checked: China GB 14930.1-2022 (the only sovereign detergent contaminant standard in the register; scope is Type A detergents used directly for food cleaning and Type B detergents for tableware and food-contact tools, equipment, packaging and containers, and it sets a total-arsenic ceiling only); China NMPA 2015 Safety and Technical Standards for Cosmetics Table 2 (lead, arsenic, mercury, cadmium); Korea MFDS Notice 2025-63 and its predecessor Notice 2024-9 (lead, cadmium, nickel, antimony, arsenic, mercury); Germany BVL 2017 technically-avoidable heavy metals in cosmetics (lead, cadmium, mercury, arsenic, antimony); Health Canada cosmetic heavy-metal impurity limits; the ASEAN Cosmetic Directive Annex (2019) and the Malaysia NPRA, Singapore HSA, Indonesia BPOM, Taiwan MoHW and New Zealand cosmetic group-standard equivalents; EAEU TR CU 009/2011; EU REACH Annex XVII entry 20 (organotin), entry 23 (cadmium in listed polymers and paints), entry 47 as amended by Regulation (EU) 301/2014 (Cr(VI) in skin-contact leather), entry 72 with Appendix 12 as amended by Regulation (EU) 2018/1513 (lead, cadmium, arsenic and Cr(VI) in consumer clothing and skin-contact textiles) and entry 75 under Regulation (EU) 2020/2081 (tattoo and permanent-make-up mixtures); EU Directive 2009/48/EC toy-safety migration limits; EU RoHS 2011/65/EU; and the dietary contaminant regimes (EU 2023/915, EU 2024/1987, Codex CXS 193, US FDA, FSANZ, Health Canada, UK, Hong Kong Cap. 132V), which are inapplicable both on scope and on basis. China NMPA 2015 Table 2, Korea MFDS Notice 2025-63, Germany BVL 2017, Health Canada, the ASEAN Annex, China GB 14930.1-2022 and EU REACH Annex XVII entry 72 Appendix 12 all omit aluminium. The only non-dietary aluminium values in the register are the EU

Total chromium (Cr)	control	No jurisdiction sets a total-chromium maximum for a laundry or fabric-care product, and no near-neighbour read-across is supportable. Instruments checked: China GB 14930.1-2022 (the only sovereign detergent contaminant standard in the register; scope is Type A detergents used directly for food cleaning and Type B detergents for tableware and food-contact tools, equipment, packaging and containers, and it sets a total-arsenic ceiling only); China NMPA 2015 Safety and Technical Standards for Cosmetics Table 2 (lead, arsenic, mercury, cadmium); Korea MFDS Notice 2025-63 and its predecessor Notice 2024-9 (lead, cadmium, nickel, antimony, arsenic, mercury); Germany BVL 2017 technically-avoidable heavy metals in cosmetics (lead, cadmium, mercury, arsenic, antimony); Health Canada cosmetic heavy-metal impurity limits; the ASEAN Cosmetic Directive Annex (2019) and the Malaysia NPRA, Singapore HSA, Indonesia BPOM, Taiwan MoHW and New Zealand cosmetic group-standard equivalents; EAEU TR CU 009/2011; EU REACH Annex XVII entry 20 (organotin), entry 23 (cadmium in listed polymers and paints), entry 47 as amended by Regulation (EU) 301/2014 (Cr(VI) in skin-contact leather), entry 72 with Appendix 12 as amended by Regulation (EU) 2018/1513 (lead, cadmium, arsenic and Cr(VI) in consumer clothing and skin-contact textiles) and entry 75 under Regulation (EU) 2020/2081 (tattoo and permanent-make-up mixtures); EU Directive 2009/48/EC toy-safety migration limits; EU RoHS 2011/65/EU; and the dietary contaminant regimes (EU 2023/915, EU 2024/1987, Codex CXS 193, US FDA, FSANZ, Health Canada, UK, Hong Kong Cap. 132V), which are inapplicable both on scope and on basis. No consumer-chemical regime in the register sets a chromium row: China NMPA 2015 Table 2, Korea MFDS Notice 2025-63, Germany BVL 2017, Health Canada, the ASEAN Annex and China GB 14930.1-2022 all omit it. The register's non-dietary

Hexavalent chromium (Cr-VI)	control	No jurisdiction sets a hexavalent-chromium maximum for a laundry or fabric-care product, and no near-neighbour read-across is supportable in this row's basis. Instruments checked: China GB 14930.1-2022 (the only sovereign detergent contaminant standard in the register; scope is Type A detergents used directly for food cleaning and Type B detergents for tableware and food-contact tools, equipment, packaging and containers, and it sets a total-arsenic ceiling only); China NMPA 2015 Safety and Technical Standards for Cosmetics Table 2 (lead, arsenic, mercury, cadmium); Korea MFDS Notice 2025-63 and its predecessor Notice 2024-9 (lead, cadmium, nickel, antimony, arsenic, mercury); Germany BVL 2017 technically-avoidable heavy metals in cosmetics (lead, cadmium, mercury, arsenic, antimony); Health Canada cosmetic heavy-metal impurity limits; the ASEAN Cosmetic Directive Annex (2019) and the Malaysia NPRA, Singapore HSA, Indonesia BPOM, Taiwan MoHW and New Zealand cosmetic group-standard equivalents; EAEU TR CU 009/2011; EU REACH Annex XVII entry 20 (organotin), entry 23 (cadmium in listed polymers and paints), entry 47 as amended by Regulation (EU) 301/2014 (Cr(VI) in skin-contact leather), entry 72 with Appendix 12 as amended by Regulation (EU) 2018/1513 (lead, cadmium, arsenic and Cr(VI) in consumer clothing and skin-contact textiles) and entry 75 under Regulation (EU) 2020/2081 (tattoo and permanent-make-up mixtures); EU Directive 2009/48/EC toy-safety migration limits; EU RoHS 2011/65/EU; and the dietary contaminant regimes (EU 2023/915, EU 2024/1987, Codex CXS 193, US FDA, FSANZ, Health Canada, UK, Hong Kong Cap. 132V), which are inapplicable both on scope and on basis. Hexavalent chromium is expressly NOT declared not material in this category, and the method-artifact reasoning that supports a not-material determination for cooked foods does not transfer: the reducing, organic-rich, wet matrix that converts Cr(VI) to

4 | Cite this standard

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References

[1] DE — human-food maximum levels
Binds Pb, Cd, tAs, tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] South Korea — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/laundry-starch-sizing-fabric-finish. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

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